

JOSHUA ERNEST PEDI REDDY

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Summary

Embedded Software Engineer with an M.S. in Robotics and 3+ years of experience developing autonomy, real-time control, state-estimation, telemetry, and embedded software for robotic and mechatronic systems. Experienced in C++, Python, ROS/ROS2, motion planning, model predictive control, EKF/UKF sensor fusion, Linux, and hardware-software integration. Proven ability to translate robotics concepts into reliable software across simulation, data analysis, edge systems, embedded platforms, and physical hardware validation.

Skills

Planning, Controls & Autonomy: Motion and Path Planning, Trajectory Optimization, Model Predictive Control, Optimal Control, Collision Avoidance, Kinematic Feasibility, Robot and Vehicle Dynamics

Localization, Estimation & Perception: State Estimation, EKF/UKF, Multi-Sensor Fusion, SLAM, LiDAR and Vision Sensor Integration, Camera Calibration, OpenCV

Robotics Software & Simulation: ROS/ROS2, C++, Python, Linux, Gazebo, RViz, MATLAB, Simulink, NumPy, SciPy, PyTorch

Embedded & Real-Time Systems: Embedded C/C++, MSP430, TI C2000/F2811, CAN, UART, Motor Control, Real-Time Scheduling, Interrupt-Driven Systems, Watchdog and Fault Handling, Sensor and Actuator Integration, Hardware-in-the-Loop and Bench Testing

Software Infrastructure & Integration: Git, Docker, Kubernetes, C#, .NET, RabbitMQ, AsyncAPI, CI/CD, Azure DevOps, Unit Testing, Integration Testing, Telemetry, Diagnostics, and Replay Workflows

PROFESSIONAL EXPERIENCE

SLB - Embedded Software Engineer | Autonomy, Controls & Production Software

Houston, TX, USA | July 2023 – Present

- Developed **Predictive Traction**, a supervisory autonomy framework for wireline robotic tractors that transforms historical job data and live telemetry into behavioral intelligence for detecting traction degradation, slip, micro-stalls, stalls, restriction interactions, and recovery events.
- Built synchronized telemetry, behavior-segmentation, replay, and diagnostic workflows across motor current, RPM, tension, acceleration, winch speed, tractor velocity, depth, CCL, arm force, and system-state signals, enabling repeatable analysis and regression evaluation.
- Defined a physics-guided state-estimation and control architecture for estimating tractor speed, slip, traction efficiency, position, and uncertainty using model-based reasoning and EKF/UKF concepts.
- Developed restriction-aware autonomy concepts for proactively adjusting tractor speed, motor torque, arm force, and traction patterns to reduce stalls, equipment stress, and non-productive operating time.
- Integrated tractor telemetry and control interfaces into the LiveOps edge platform using C#/.NET, RabbitMQ, AsyncAPI, Docker, and Kubernetes; implemented tool-specific data-stream registration, telemetry publishing, and automated integration tests.
- Developed embedded firmware for MSP430-based systems, including dual MEMS sensing, CAN communication, non-blocking motor speed and ramp control, modular task scheduling, and sensor and actuator integration.
- Brought up and debugged TI C2000/F2811 real-time firmware, resolving timer and interrupt failures, watchdog configuration issues, stack corruption, and blocking CAN service loops while adding timeout and fault-handling mechanisms.
- Improved software reliability through code reviews, unit and integration testing, hardware bench validation, telemetry replay, Azure DevOps CI/CD pipelines, SonarQube quality gates, and system-level failure analysis.

XLAB for Autonomous Systems, University of Pennsylvania - Robotics Research Assistant

Philadelphia, PA | May 2022 – Oct 2022

- Implemented differentiable trajectory-generation frameworks for car-like robots using interpolating radial basis function networks to approximate parameterized optimal-control solutions.
- Evaluated trajectory cost, kinematic feasibility, and environment-dependent collision risk over discretized grids to support computationally efficient autonomous-vehicle motion generation.
- Developed lower-latency trajectory and cost-evaluation methods intended to reduce dependence on computationally expensive online optimization.

Maverick Medicare Systems - Team Lead

Hyderabad, India | 2019 – 2021

- Co-founded and led a technology startup developing a smart healthcare decision-support platform, owning product strategy, prototype development, backend API architecture, customer discovery, and go-to-market iteration.
- Directed a 20+ member cross-functional team across software, product, and operations while building award-winning prototypes recognized through national innovation awards, AICTE/MHRD-supported funding, international exhibitions, and design patents. Delivered real-time decision-support workflows for single and multi-user scenarios, reducing prototype failure rates below 5% through iterative testing, customer feedback, and system-level refinement.

PROJECTS & ACHIEVEMENTS

F1TENTH Head-to-Head Autonomous Racing

University of Pennsylvania | 2022

- Developed a closed-loop autonomous racing stack for an Ackermann-steered vehicle using C++, Python, ROS, LiDAR, Intel RealSense, and NVIDIA Jetson Xavier NX.
- Implemented RRT-based motion planning, collision checking, local trajectory selection, and reactive overtaking and blocking behaviors; secured second place in the ICRA 2022 F1TENTH head-to-head racing competition.
- Built a two-stage EKF/UKF pipeline to estimate vehicle states and tire-road friction coefficients by combining noisy measurements with vehicle-dynamics model predictions.
- Evaluated friction-aware estimation as a method for reducing simulation-to-real-world trajectory-tracking mismatch under changing surface conditions.

Franka Emika Panda Motion-Planning Challenge

University of Pennsylvania | 2022

- Implemented forward and inverse kinematics, collision checking, obstacle avoidance, and discrete path planning for safe manipulation of static and moving objects. Delivered the winning solution for the University of Pennsylvania manipulation challenge, later demonstrated at ICRA 2022.

Model Predictive Control for 7-DOF Manipulator Motion

University of Pennsylvania | 2022

- Developed an online model predictive planning and control framework for a 7-DOF manipulator operating around static and moving obstacles.
- Formulated constrained trajectory-tracking objectives incorporating robot dynamics, collision avoidance, actuator limits, control effort, and terminal accuracy.
- Evaluated closed-loop performance across environments with varying obstacle configurations and task complexity.

EDUCATION

University of Pennsylvania - M.S. in Robotics | GPA: 3.71/4.0

Philadelphia, PA | Aug 2021 – May 2023

- Relevant Coursework: Controls & Optimization with Applications in Robotics, Autonomous Racing, Learning in Robotics, Introduction to Robotics, Advanced Machine Perception, Computer Vision, Machine Learning

Sreenidhi Institute of Science and Technology - B.Tech. Electronics & Communication Engineering | GPA: 9.53/10.0

Hyderabad, India | Jul 2016 – May 2020

- Publications: First author on peer-reviewed research in [AI-IoT medical expert systems \(ICOEI\)](#) and [smart assistive devices \(ICACII\)](#).